

Facilitated Positional Release

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OPP3

PowerPoint and lecture built upon
the previous work of many faculty,
most recently,
Jordan Keys D.O.

Objectives:

- 1) Recognize the **definition, origins of FPR** and descriptions of **modifications** to the technique that have occurred over time
- 2) Identify the **mechanisms** thought to underlie the effectiveness of the technique – including muscle spindle gamma loop activity
- 3) Recognize **benefits, indications, and contraindications** for the application of FPR
- 4) Recognize a correct description of the **steps for treatment** using Facilitated Positional Release (FPR) in the classic teaching model
- 5) Identify key concepts related to **facet orientation** that impact the process of attaining a neutral position in the cervical, thoracic, and lumbar spine
- 6) Identify key **differences** between **FPR and Still Technique**
- 7) Identify specific **anatomic factors** to consider in the attainment of a position of ease in all available planes of motion when applying FPR treatment to the **cervical, thoracic, and lumbar spine** in multiple treatment positions

Facilitated Positional Release - Definition

“A system of **indirect myofascial release** treatment. The component region of the body is placed into a **neutral position**, diminishing tissue and joint tension in all planes, and an **activating force** (compression or torsion) is added. A technique developed by Stanley Schiowitz, D.O.”

--Glossary of Osteopathic Terminology

Facilitated Positional Release - Definition

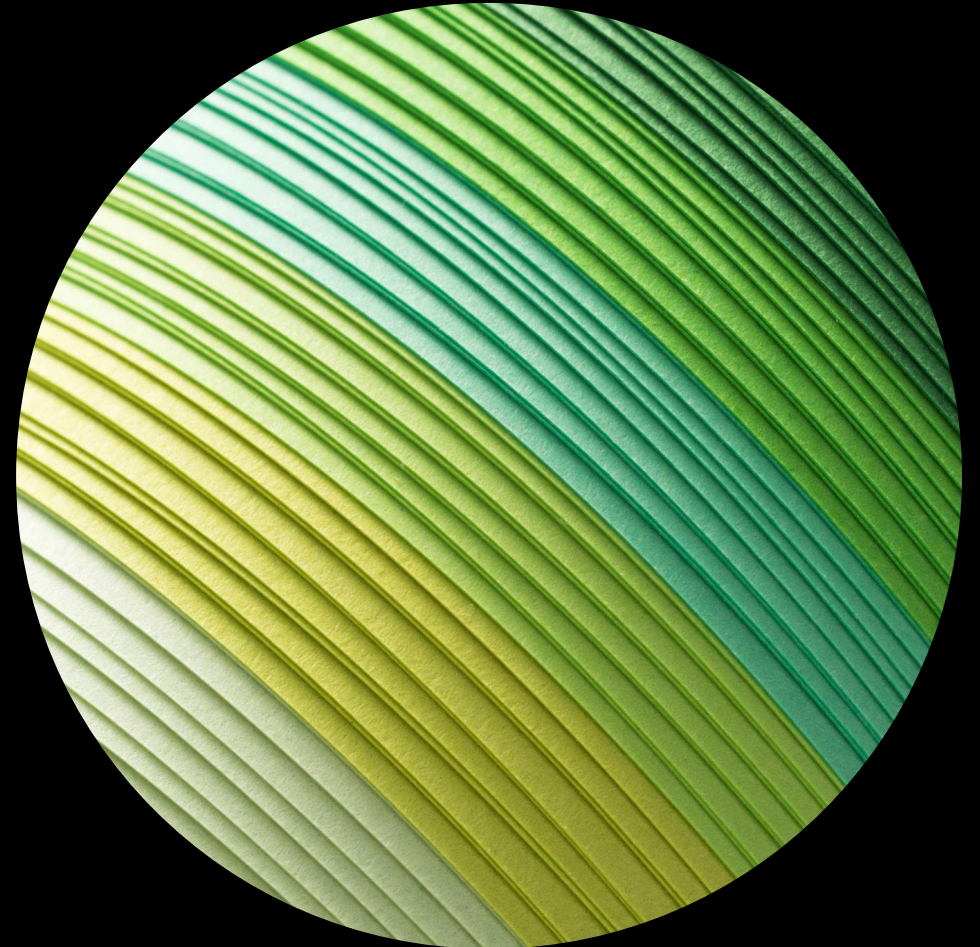
Considered to be a technique related to Myofascial Release (MFR)

Applied in a classic approach as an indirect technique, starting with the spine in a neutral position

Uses a facilitating force, usually compression but sometimes traction or torsion

Directed to the muscles involved in somatic dysfunction, shortening the superficial muscle or placing the segmental somatic dysfunction in the position of ease to affect the deep muscles.

Has been adapted for the ribs and extremities.



Facilitated Positional Release - Background

Developed by **Stanley Schiowitz, DO, FAAO**

- 1944 Graduate of the Philadelphia College of Osteopathic Medicine
- Clinical practice for 30+ years in Brooklyn, NY
- Participated in founding the New York College of Osteopathic Medicine (NYCOM) in the 1970s, eventually became Associate Dean and Chair of the Osteopathic Principles and Practices (OPP) Department in 1981 and Dean of the school in 1992
- Busy physician teaching undergraduate fellows and treating patients – waiting 90 seconds for Counterstrain was “too long”
- In 1980s, started applying a technique noted by the fellows but not initially fully described – they would chart as “Instant Osteopathy (IO)”

Facilitated Positional Release - Background

Developed by Stanley Schiowitz, DO, FAAO

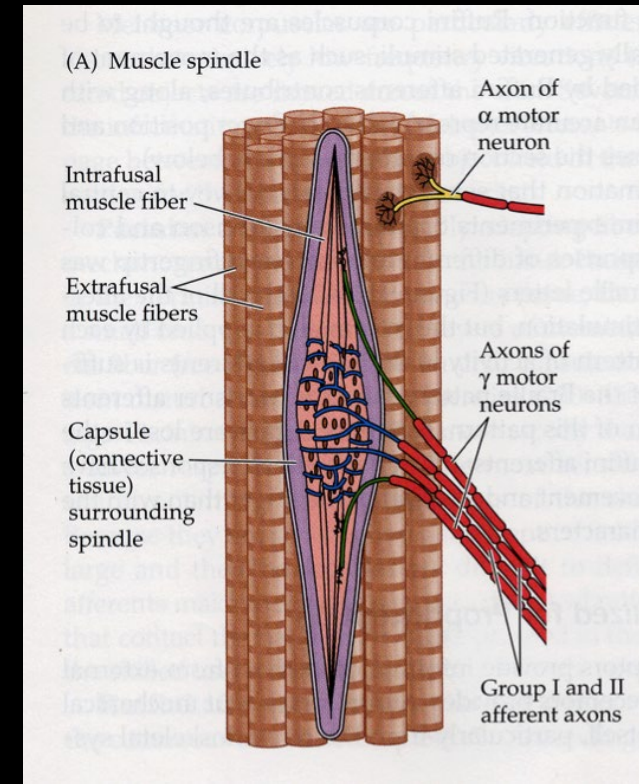
- FPR was eventually described in NYCOM teaching manuals and then published in “An Osteopathic Approach to Diagnosis and Treatment”
- Further developed in Foundations of Osteopathic Medicine and materials used for teaching practicing physicians in seminars by Dr. Schiowitz, starting in the 1990s
- Continued seminars taught by proteges of Dr. Schiowitz after his death in 2011

Neuromuscular Reflexes

- Protect Stability of Muscle.
- **Muscle Spindle:**
 - **Muscle Length.**
 - **Rate of change in muscle length.**
 - **Protect muscle from overstretching (muscle tearing).**
- **Golgi Tendon:**
 - **Muscle Tension.**
 - **Rate of change in muscle tension.**
 - **Protects attachment portion of muscle (tearing off bone/avulsion).**

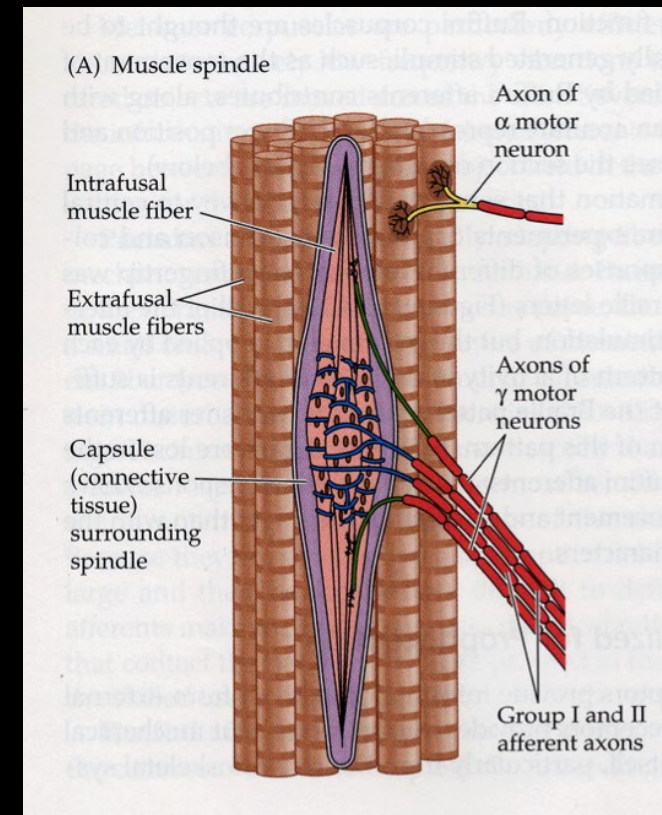
Muscle Spindle Reflex

- **Intrafusal** Mechanoreceptors within muscle belly of skeletal muscles.
- **Parallel to extrafusal skeletal muscle fibers.**
- **Intrafusal Fibers :**
 - Nuclear Chain (2-5)
 - Nuclear Bag (6-10)
 - Central portion has poor contractility; ends have greater contractility



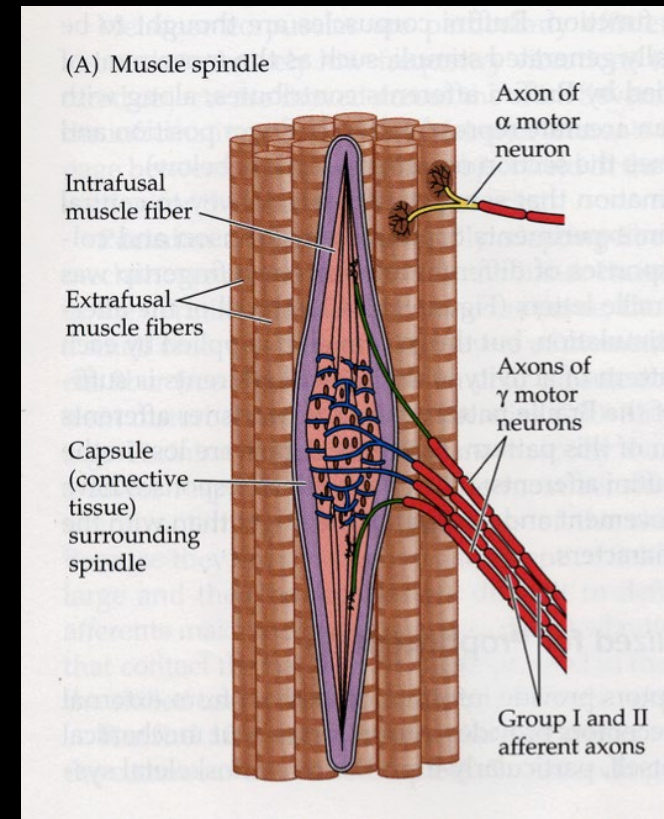
Muscle Spindle Reflex

- **Sensory Afferents:**
 - **Ia** (primary endings):
 - Innervate **every intrafusal** fiber.
 - Surround the center of the spindle like a coil (*annulospiral endings*).
 - **Rate of change.**
 - **II** (secondary endings):
 - **Nuclear chain endings only.**
 - Terminate on either side of Ia fibers in branch-like fashion (*flower-spray endings*).
 - **Length of change.**



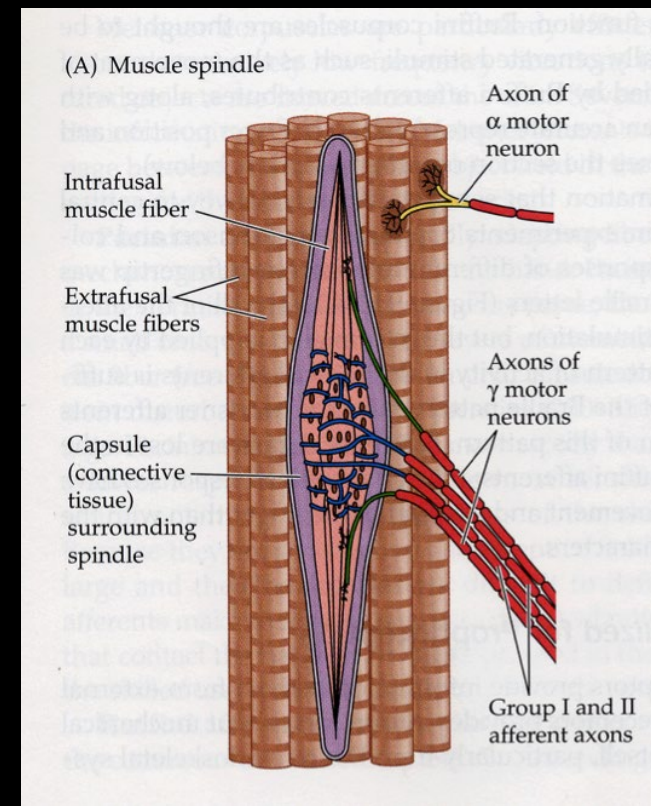
Muscle Spindle Reflex

- Motor Efferents:
 - Alpha motor neurons:
 - Innervate **extrafusal** fibers.
 - Results in contraction of shortening of whole muscle unit.



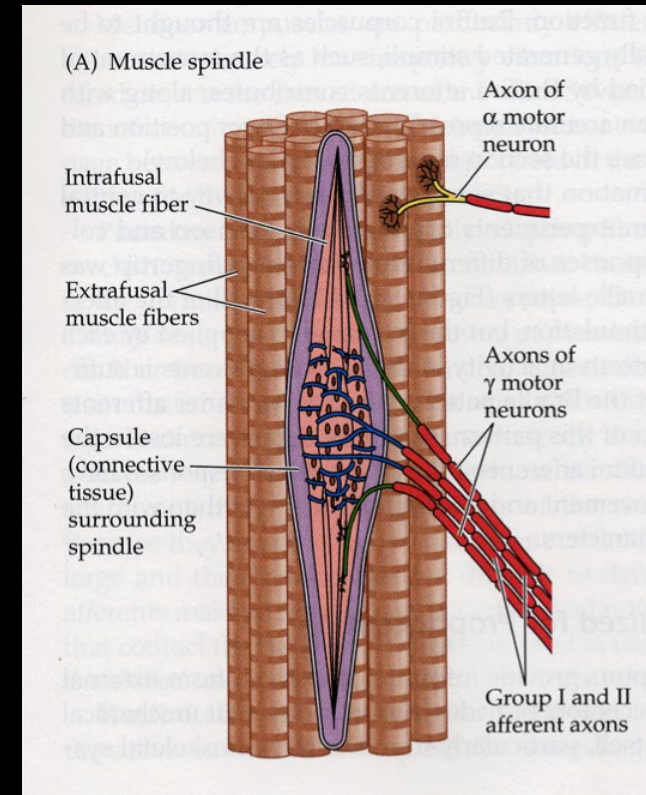
Muscle Spindle Reflex

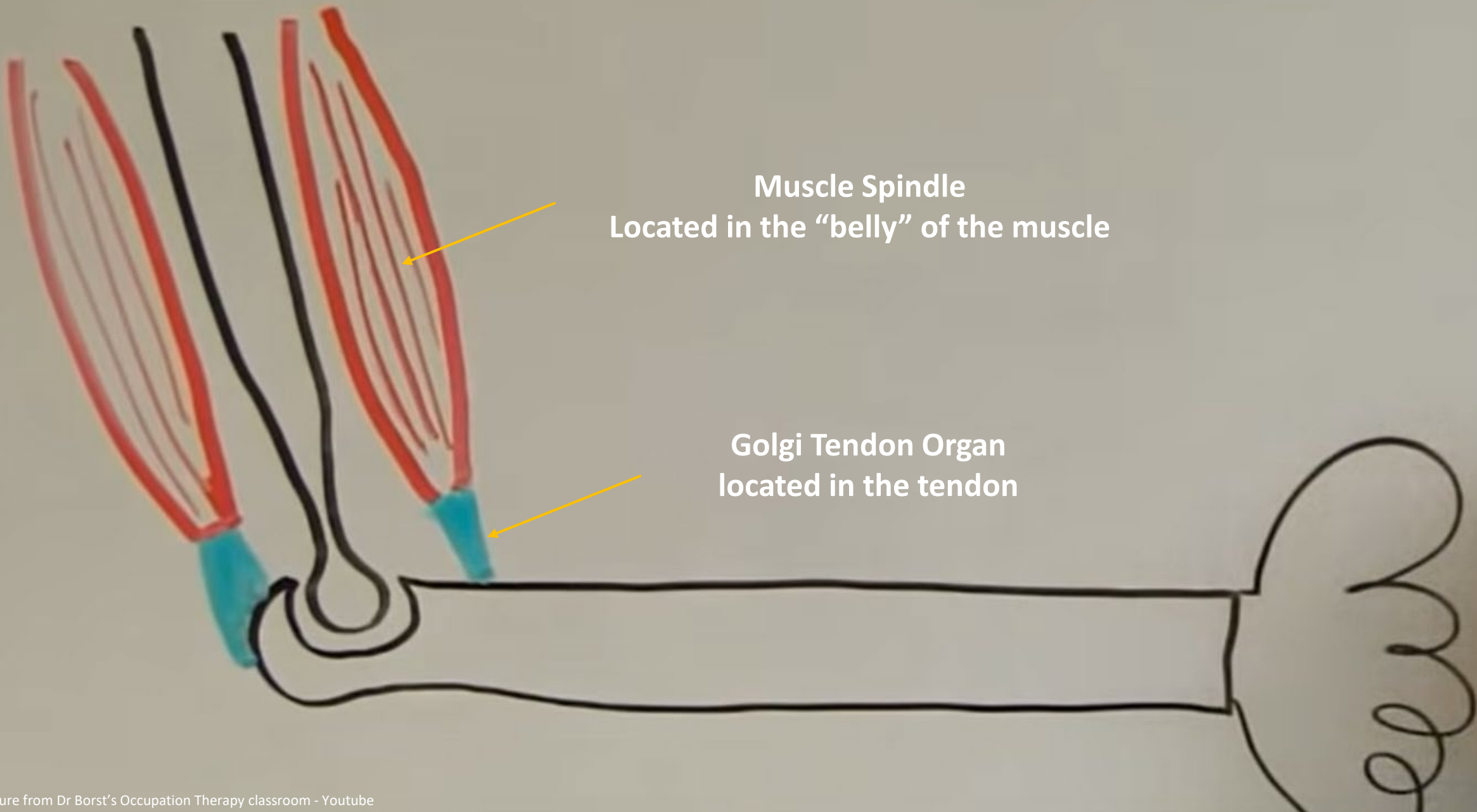
- **Motor Efferents:**
 - **Gamma Motor Neurons**
(Contraction secondary to Ia and II fibers)
 - **Maintenance of steady state**
vulnerable to overregulation.
 - Sudden stretch can result in powerful muscle contraction (inappropriate stimulation of gamma motor neurons)
 - Contractions increases sensitivity of sensory fibers.



Muscle Spindle Reflex

- Stimulation occurs with stretching of midpoint:
 - Lengthening of muscle.
 - Contraction of endpoints.
- Stretching MS increases firing and shortening MS decreases firing.
- Neutral position=baseline firing.
- Hypershortened position=cessation of firing.
- Activation of muscle spindle results in overall contraction of muscle.





A hand-drawn diagram on a light-colored background. It depicts a muscle and its attachment to a bone. The muscle is shown as two red, elongated, striated structures. A yellow arrow points from the text 'Muscle Spindle' to one of these muscle structures. The muscle is attached to a thick black line representing a tendon. A yellow arrow points from the text 'Golgi Tendon Organ located in the tendon' to a small blue, cylindrical structure on the tendon. The tendon is connected to a larger, rounded structure on the right, which is also drawn in black ink.

Muscle Spindle
Located in the “belly” of the muscle

Golgi Tendon Organ
located in the tendon

Facilitated Positional Release – Benefits and Indications

- **Rapid, non-traumatic** treatment to an affected area
- **Easy application of further treatment techniques**
- Can be **utilized in any part of a treatment sequence**:
 - Initial treatment
 - Subsequent treatment following another technique such as Soft Tissue
 - Set-up for other techniques such as Muscle Energy or HVLA
 - Repeat treatment if initial response not adequate
- Treatment in most regions **can be applied in multiple treatment positions** - seated, supine, prone, or side-lying position

Facilitated Positional Release - Contraindications

- Few contraindications and MOST are relative
- Caution:
 - Osteoporosis, malignancy, congenital malformations, stenosis, rheumatological disorders
- Relative (if patient doesn't tolerated position):
 - Cervical disc herniation with radiculopathy (if present a cervical spine compression can mimic a Spurling maneuver, producing symptoms of cervical radiculopathy - you can use traction instead of compression)
 - Lumbar discogenic pain (may need to modify the technique)
 - Any recent surgery such as a hip replacement (external rotation and torque may risk disarticulation of the joint)
 - First Rib and shoulder pathology (may need to adjust position)
- Absolute:
 - Fracture in the region being treated (compression can change stable fracture to unstable).
 - Absence of somatic dysfunction



Facilitated Positional Release – Basic Treatment Sequence

- 1) The physician modifies the patient's **sagittal posture** in the region to be treated (**Neutralizes- Idling of facet joints**)
- 2) A **facilitating force** is applied*
- 3) The **large muscle is shortened** or the somatic dysfunction is placed into its **freedoms of motion (ease)**
- 4) The position is held for **3 to 5 seconds**
- 5) Release the position and re-evaluate

* In some cases, the position of ease is attained before the facilitating force is applied

Facilitated Positional Release Post Treatment

- Patients should be given **usual post-treatment instructions** for possible transient fatigue or discomfort related to restoration of circulation to previously dysfunctional tissues

Facilitated Positional Release – Treatment Sequence Modifications

As described in *Foundations of Osteopathic Medicine*:

- The basic steps are applied, and an initial release achieved
- There may be the addition of a direct component of treatment, moving in the barrier directions. This is similar to application of the Still Technique as described by van Buskirk.

As taught by Dr. Schiowitz in courses:

- The addition of “jiggling,” a type of micro-articulatory oscillation directed to the segment being treated

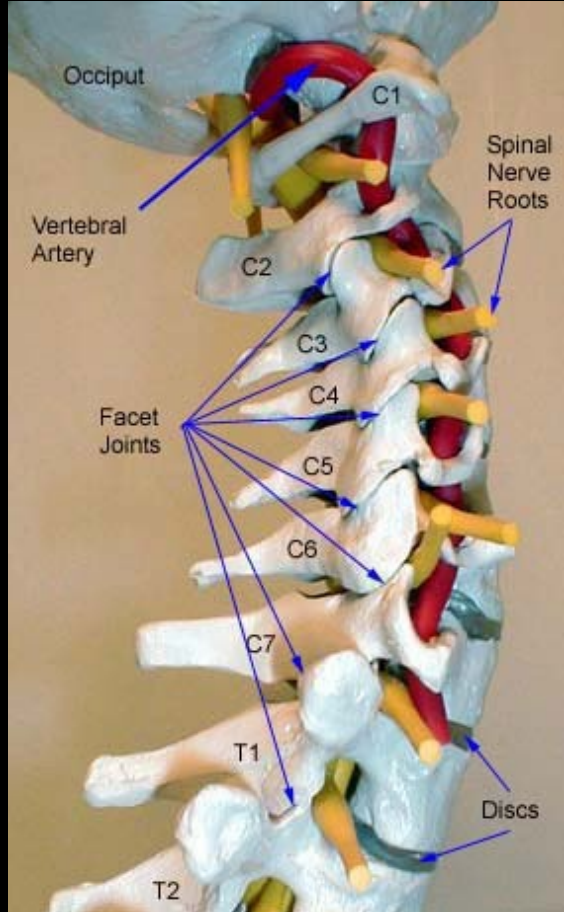
Still Technique

- Very Similar to FPR but the **difference is continued movement.**
- Carl Phillip McConnell, D.O., Herb Yates D.O., PhD, Richard Van Buskirk D.O., Ph.D, Edward Stiles, D.O. (Still-Laughlan Technique).
- **Indirect then direct**
- Procedure:
 - Evaluate the affected area and place into a **position of ease**
 - **Introduce a force vector** to the affected tissue from another part of the body. This compression or traction should be less than 5 lbs.
 - **Use the force vector to move** the affected tissue initially **into a smooth path** to its position of ease **and then** alter the position while **maintaining the force toward and through** to the **position of the restriction** of position and movement.
 - As the tissue moves through its restriction a “bump” and or “click” may be felt or heard. Neither is necessary for correction of the somatic dysfunction.
 - The region is **passively moved back to neutral** and **retested**.

Axial Skeleton

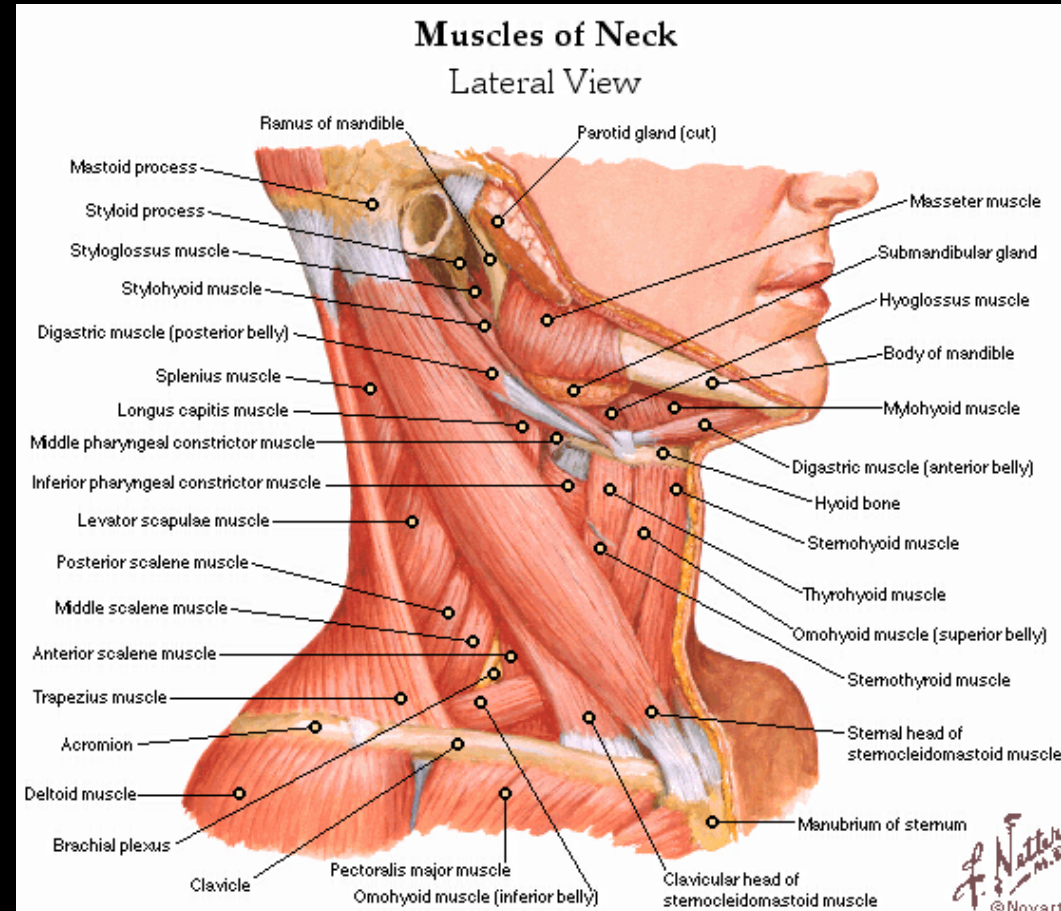


Facilitated Positional Release – Achieving a Neutral Spine Position Cervical Spine

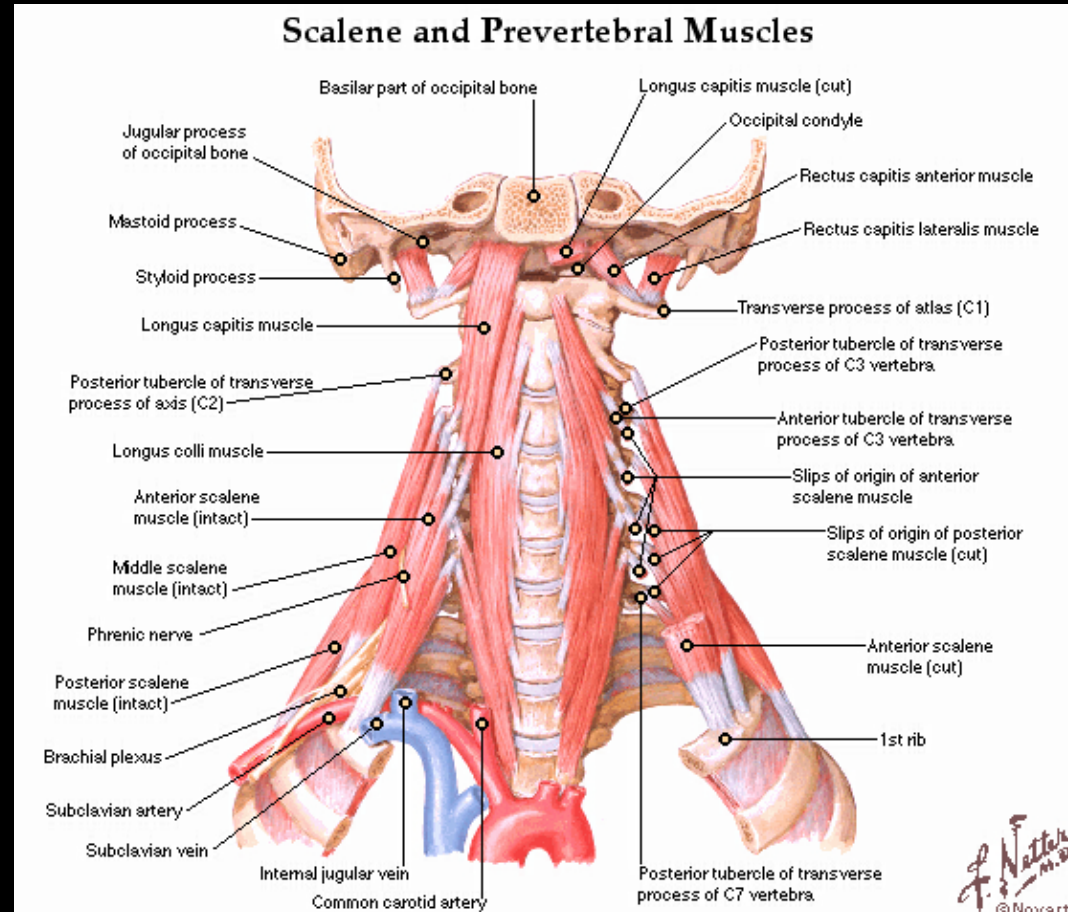


- Orientation of facet joints determines proper positioning to achieve neutral spine between flexion and extension
- Neutral cervical spine position usually involves a reduction of normal lordosis

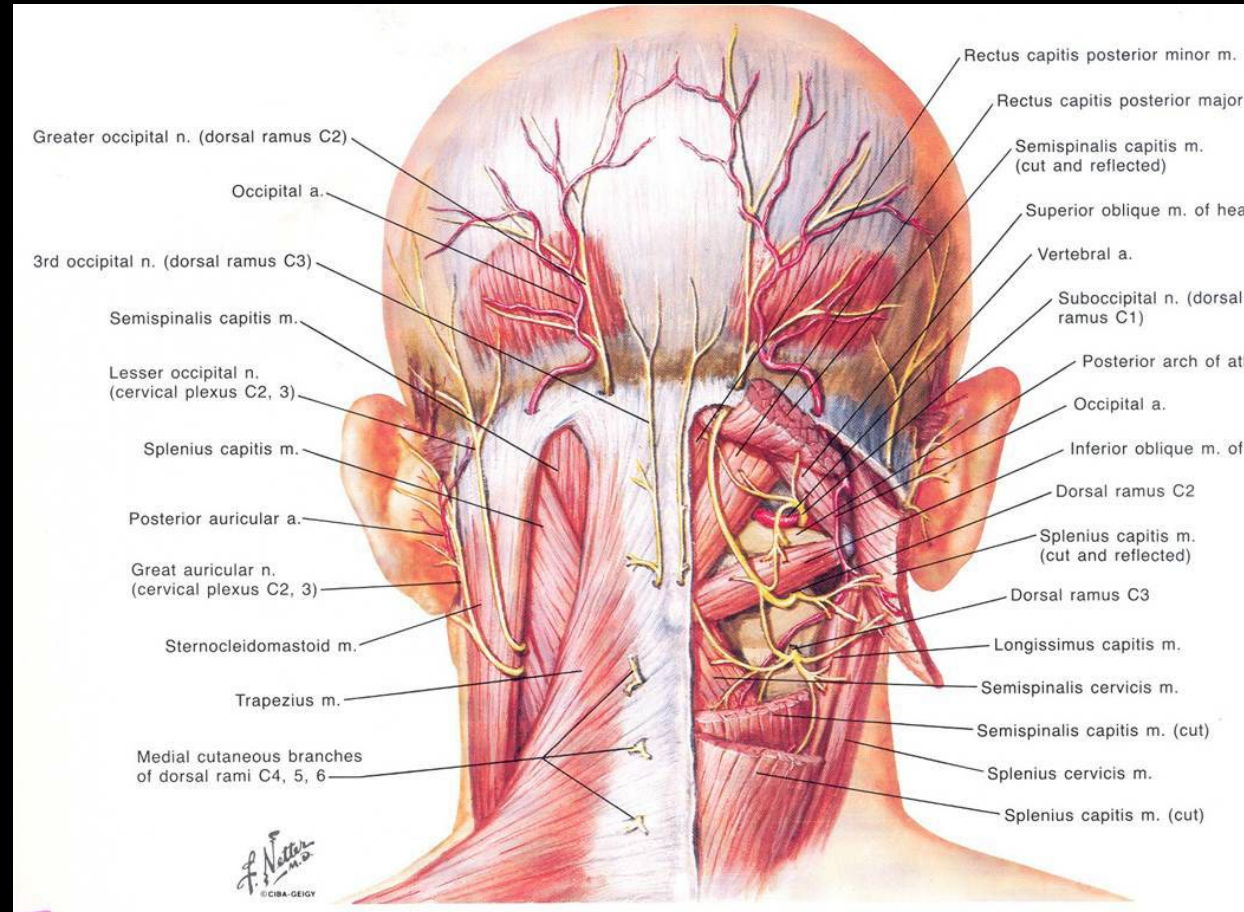
Cervical Muscles - Superficial



Cervical Muscles - Deep



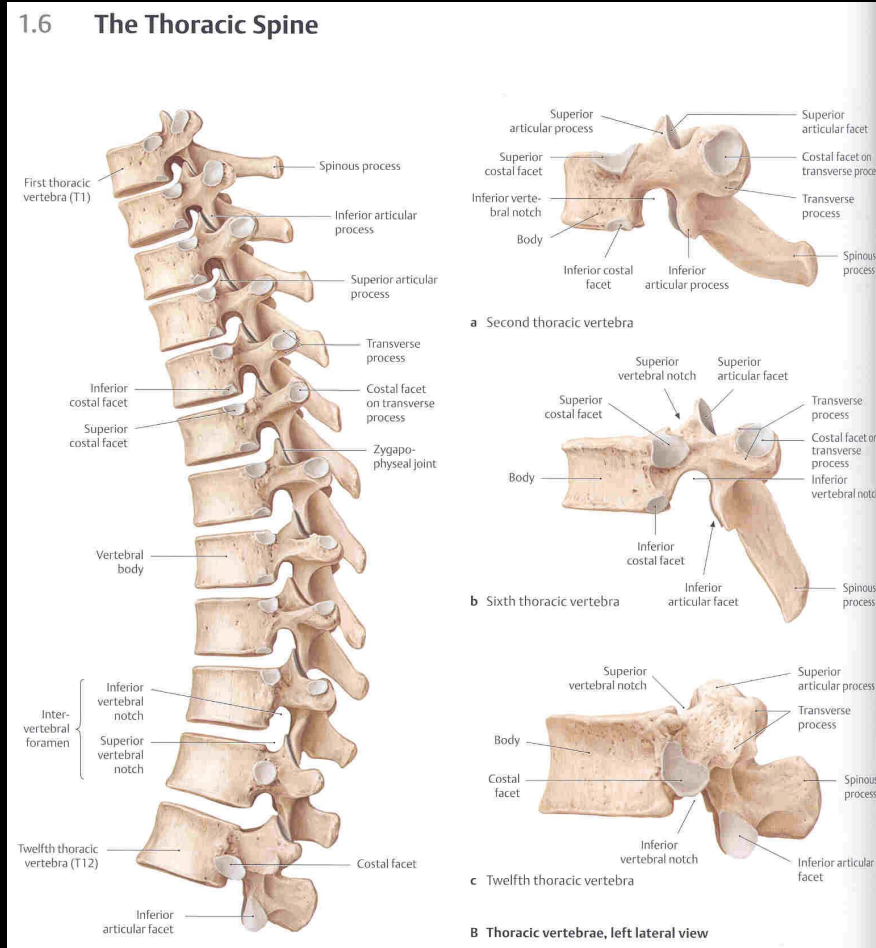
Suboccipital Triangle





Cervical
Spine
FPR

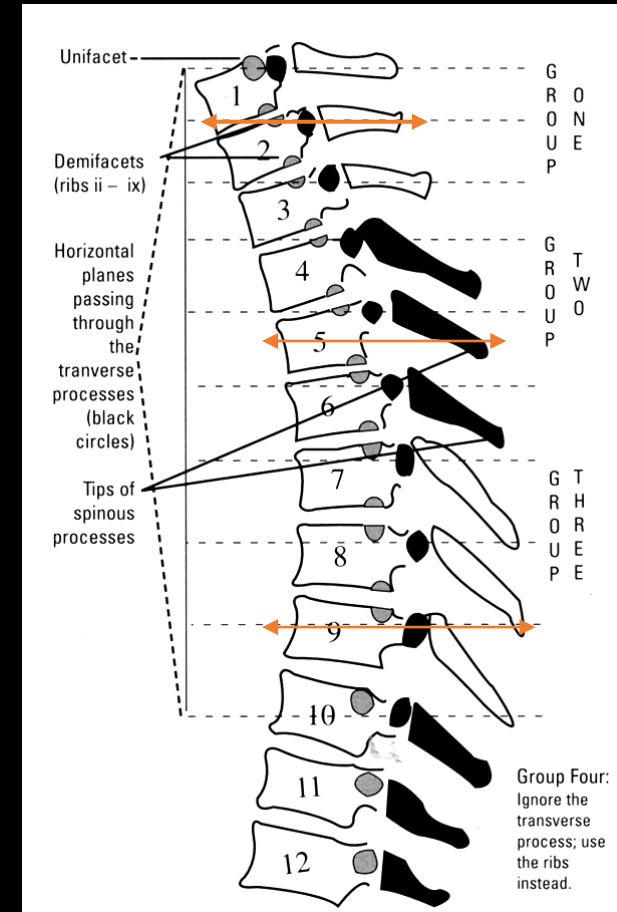
Thoracic Spine



- Orientation of facet joints determines proper positioning to achieve neutral spine between flexion and extension
- Neutral thoracic spine position usually involves a reduction of normal kyphosis

Rule of Threes for Thoracic Vertebrae

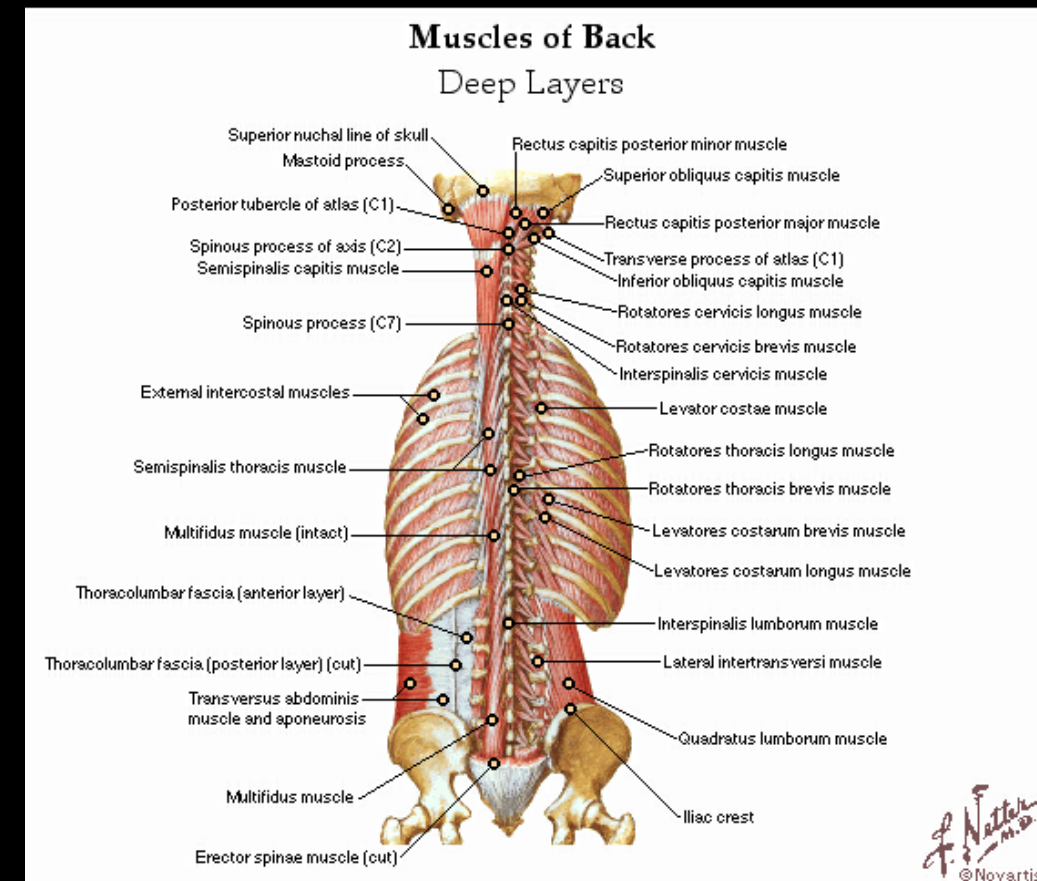
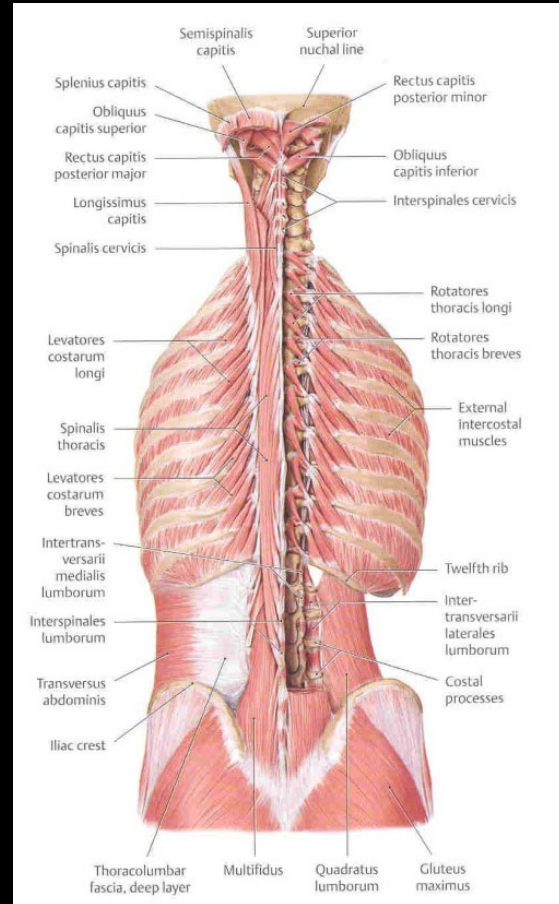
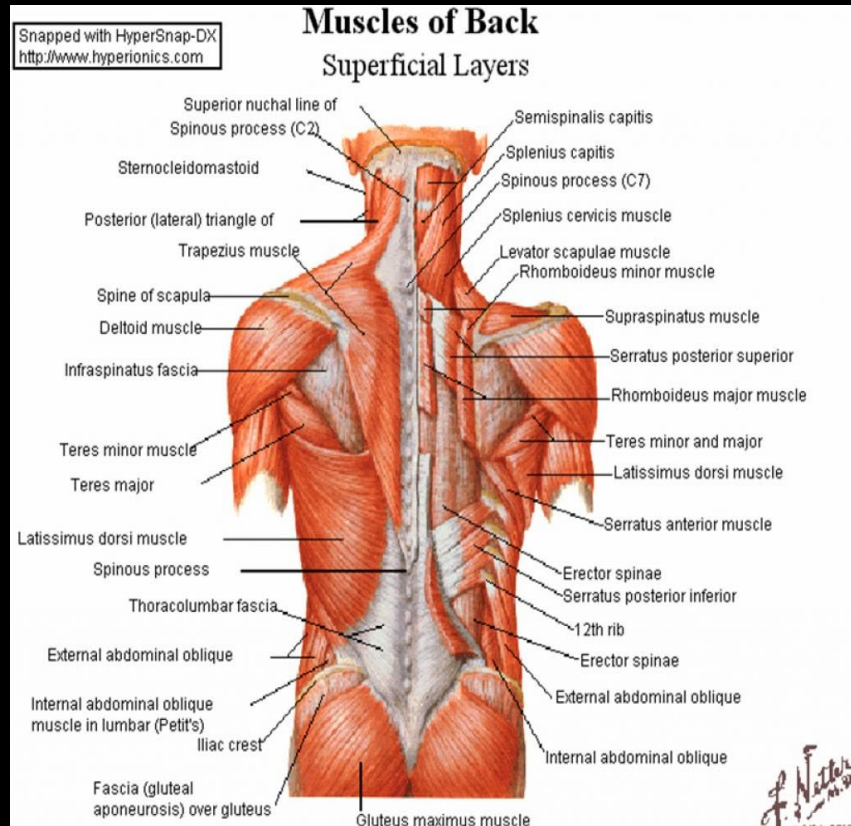
- Relationship spinous process (SP) to transverse processes (TP):
 - **T1-T3**
 - SP about **level with the TP**
 - **T4-T6**
 - SP about **½ vertebral level below** corresponding TP
 - **T7-T9**
 - SP **one level below** the TP
 - **T10**
 - **Same as T7-9** - SP one level below the TP
 - **T11**
 - **Same as T4-6** - SP about ½ vertebral level below corresponding TP
 - **T12**
 - **Same as T1-3** - SP about level with the TP





Dr Moloney's Rule:
Use a Known to
find and Unknown

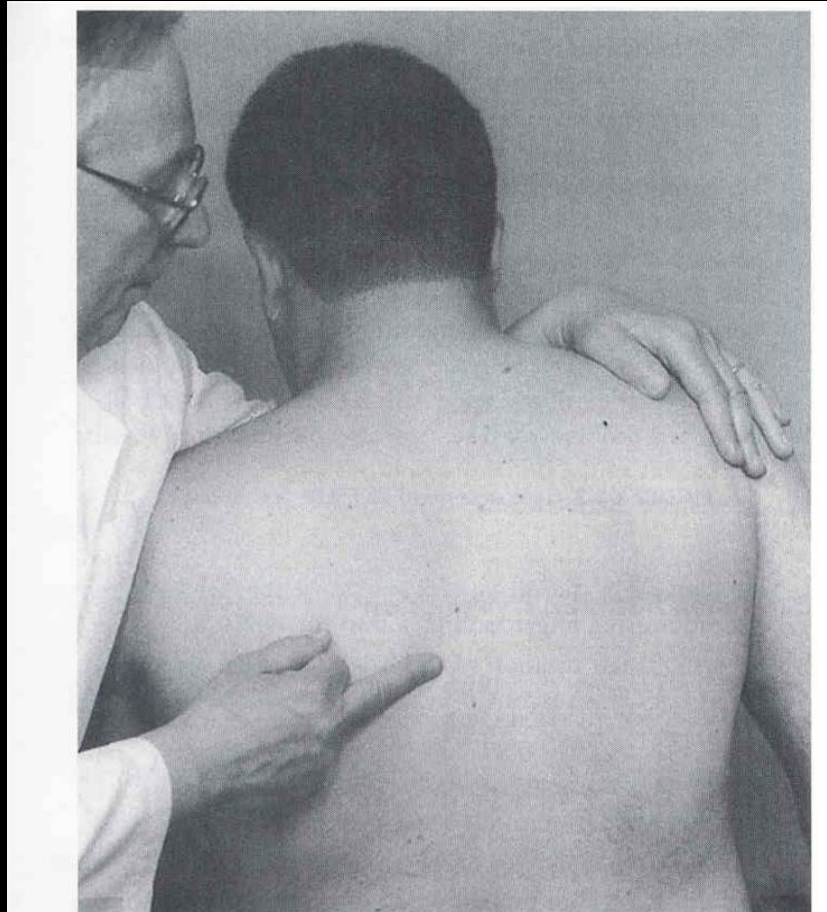
Muscles of the Back



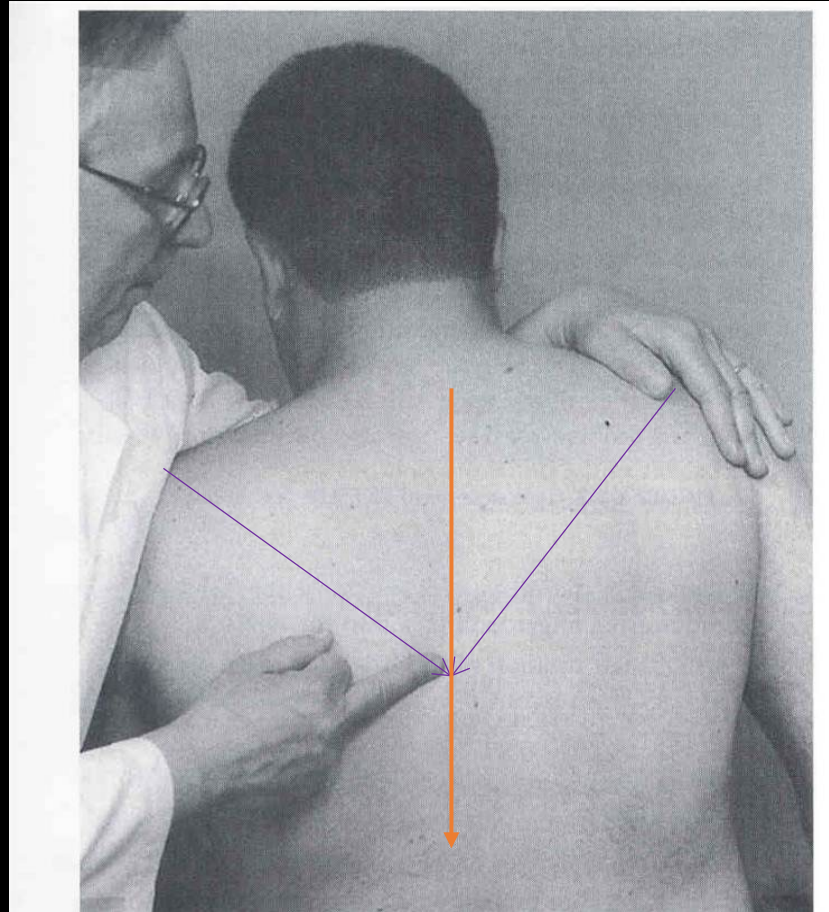
Achievement of a Neutral Spine Position

- Although FPR is a **passive** technique, it **may be necessary for the patient to initiate postural positions** in order to achieve a neutral spine position at the beginning of the treatment sequence
- The patient can be instructed to **“sit up straight”**, lift the sternum, or lift the shoulders to achieve a neutral spine
- Monitor by response of the facets to a neutral position, neither open nor closed

FPR Thoracic Spine – Seated
T6 FR_LS_L

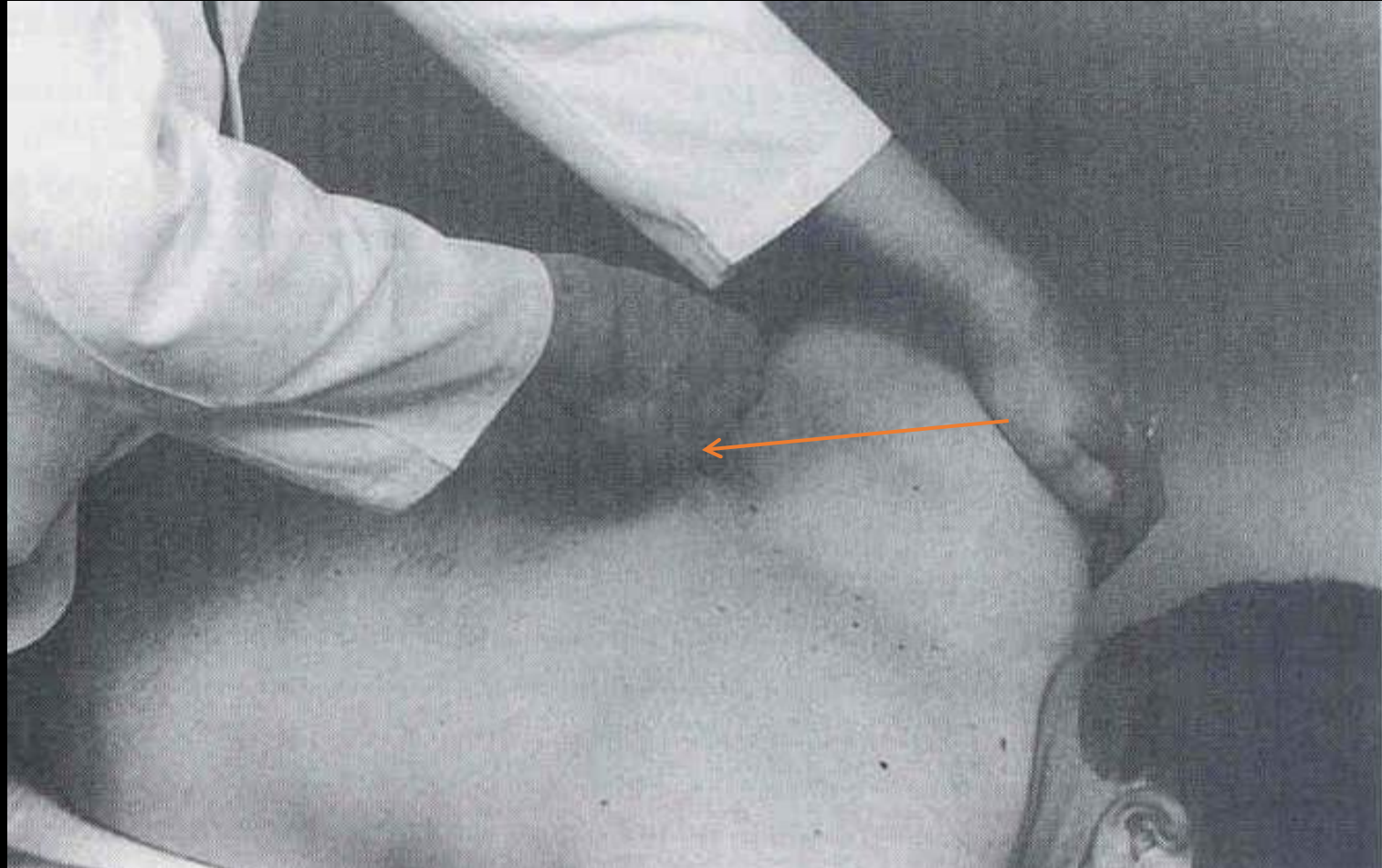


FPR Thoracic Spine – Seated Vector for compressive force



FPR Thoracic Spine – Prone

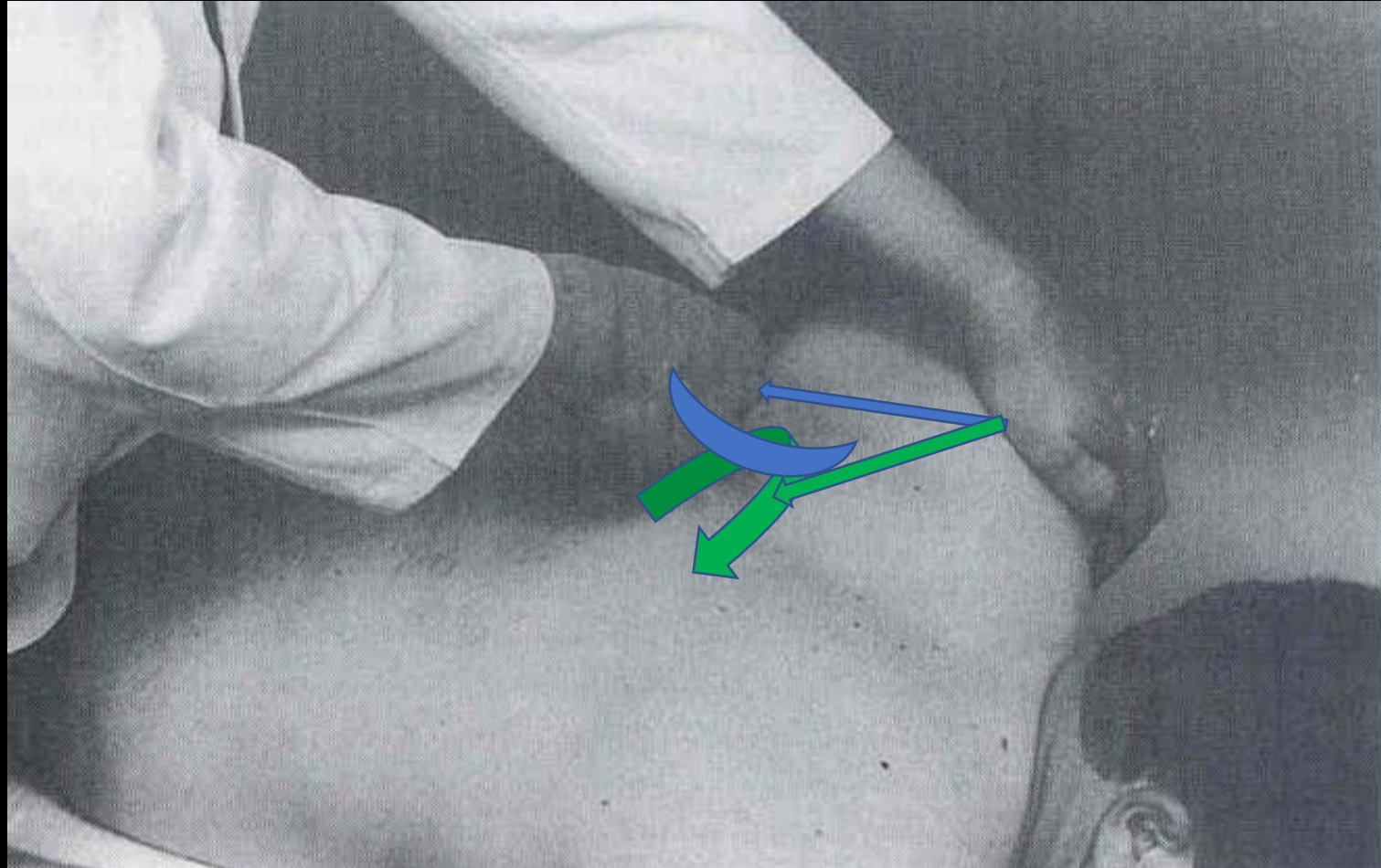
T6 ER_LS_L – compressive force



Note the photograph is mislabeled in FOM 3&4 as ESRRR, but the text is correct in the as ESLRL

Foundations 4

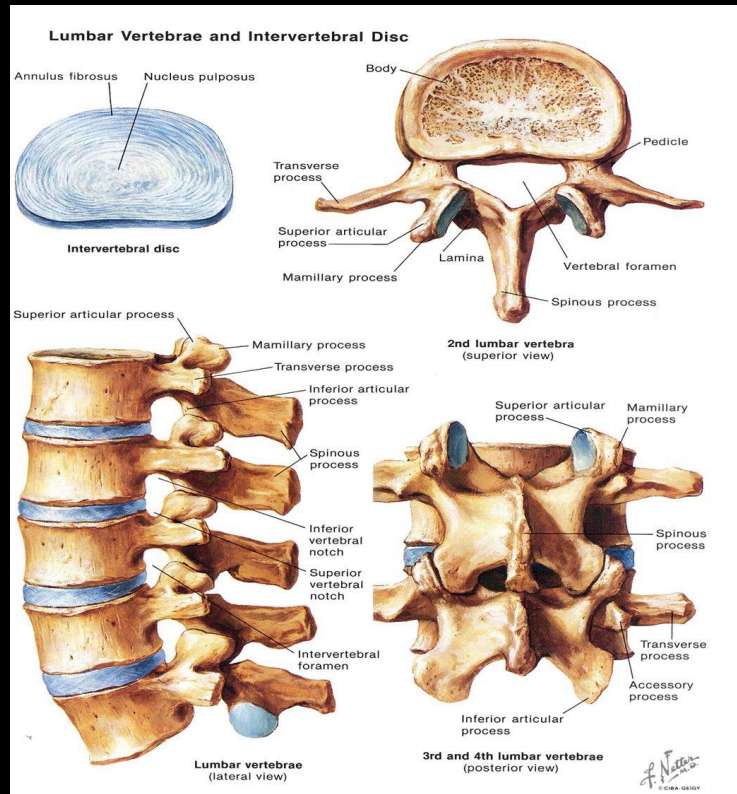
FPR Thoracic Spine – Prone T6 ER_LS_L – directions of ease



Note the **photograph** is mislabeled in FOM 3&4 as **ESRRR**,
but **the text** is correct in the as **ESLRL**

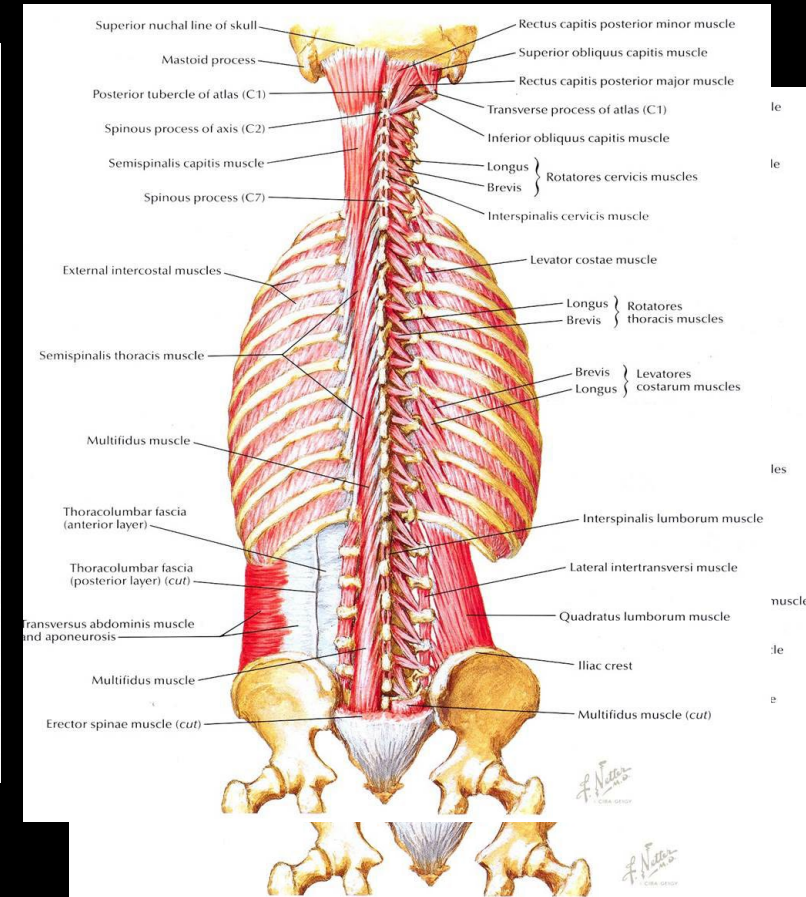
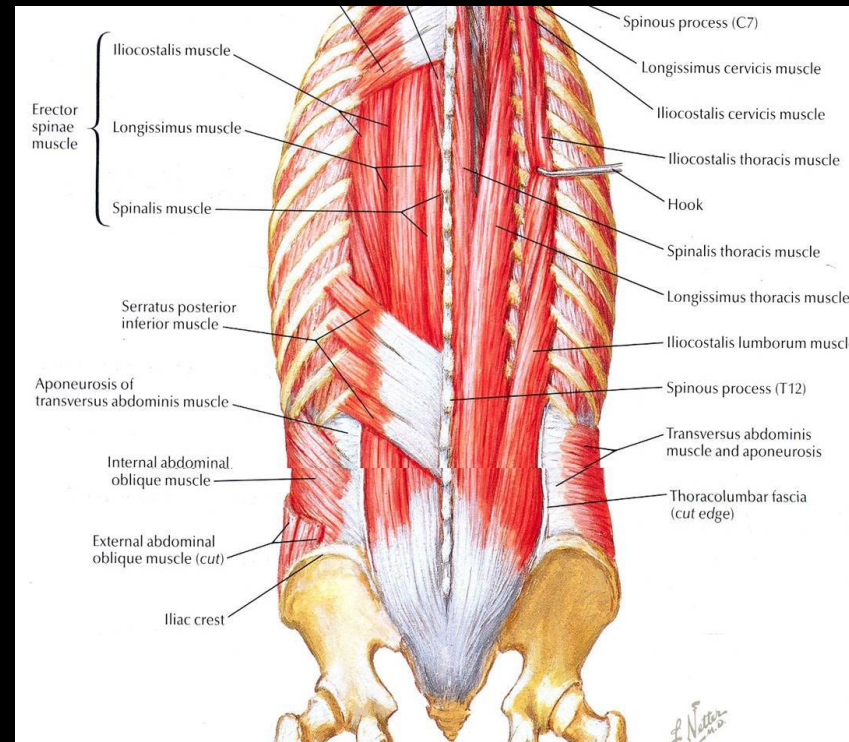
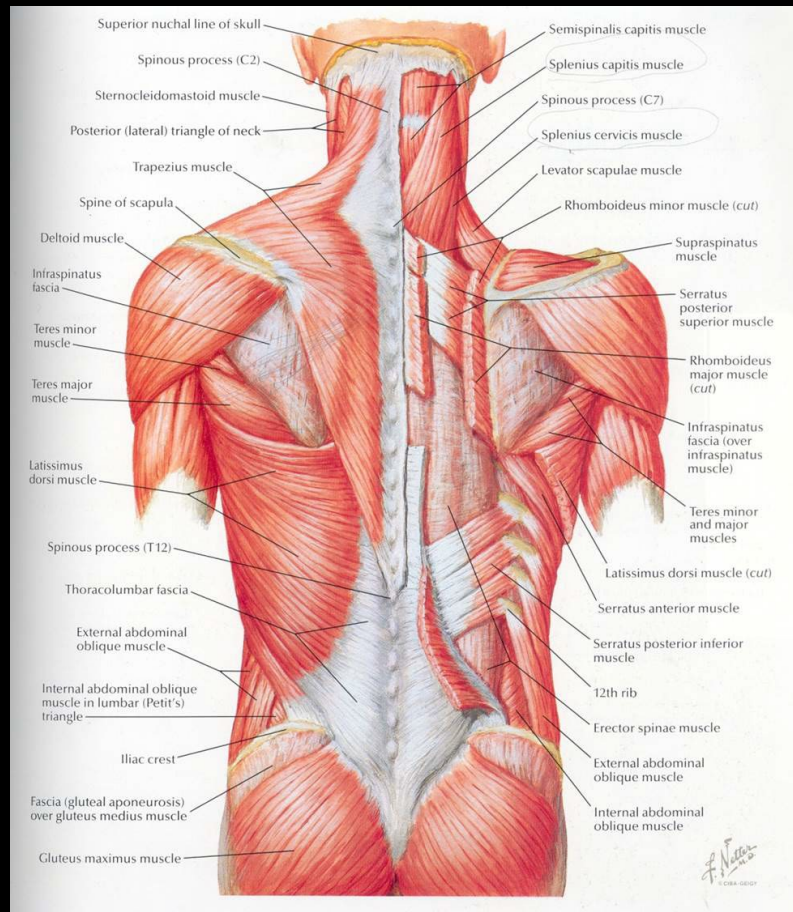
Foundations 4

Lumbar Spine – Anatomy



- Orientation of facet joints determines proper positioning to achieve neutral spine between flexion and extension
- Neutral lumbar spine position usually involves a reduction of normal lordosis

Lumbar muscles



FPR Lumbar Spine Lateral Recumbent

- **Positioning** is **opposite** to that used for **Muscle Energy Technique** – place the side of the **posterior transverse process up** (away from the table)
- MET and HVLA engage the direct barriers
- **FPR** positions into the **ease** in all planes of motion
- Benefit of lateral recumbent position is the ability to treat all planes of motion without requiring that patient be able to lay in a prone position
- Avoids the need for pillows to produce a neutral spine position at the beginning of treatment



L2 FR_LS_L

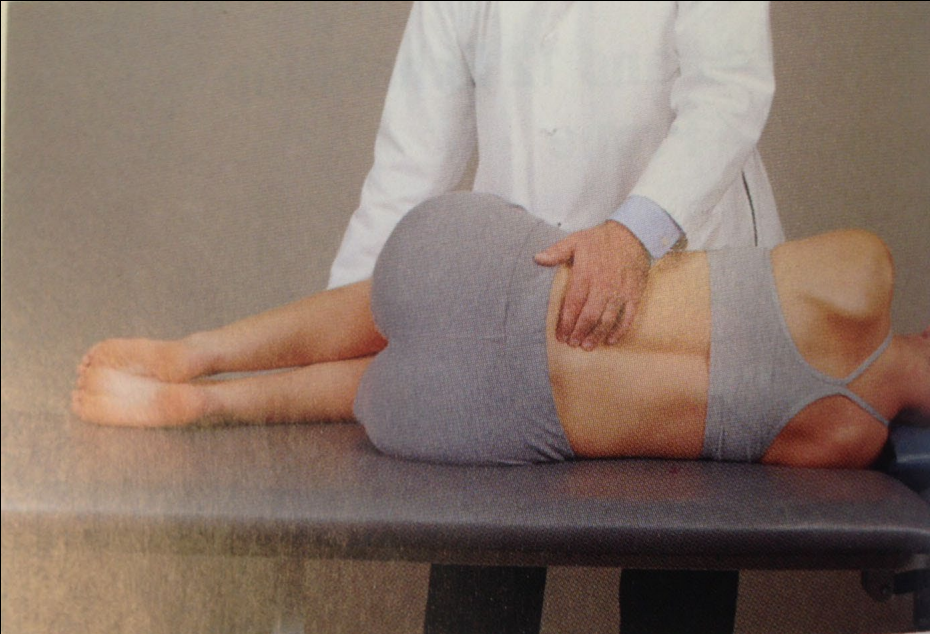


FIGURE 11.90.

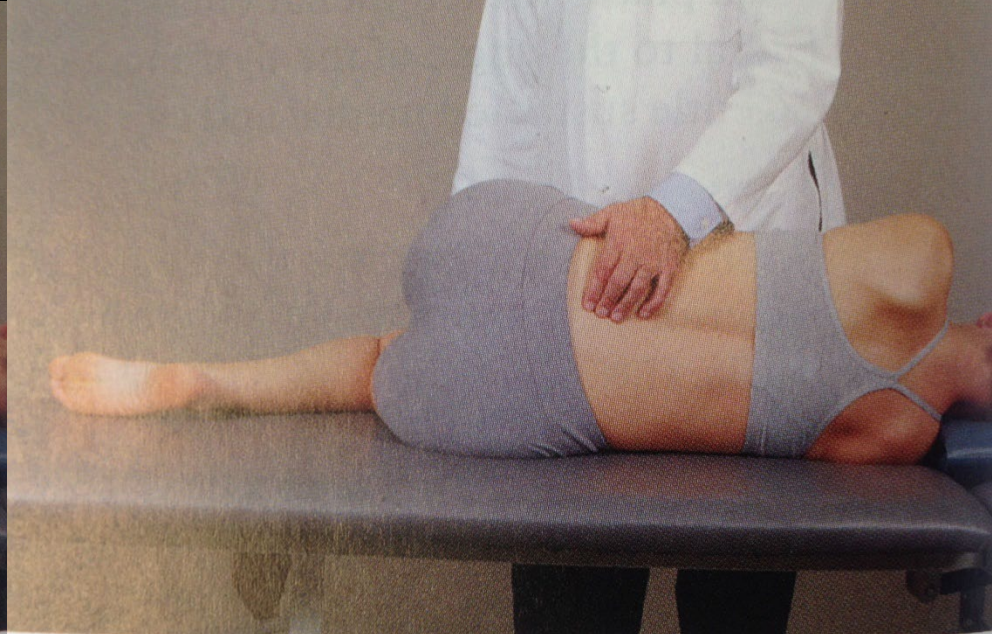


FIGURE 11.91.

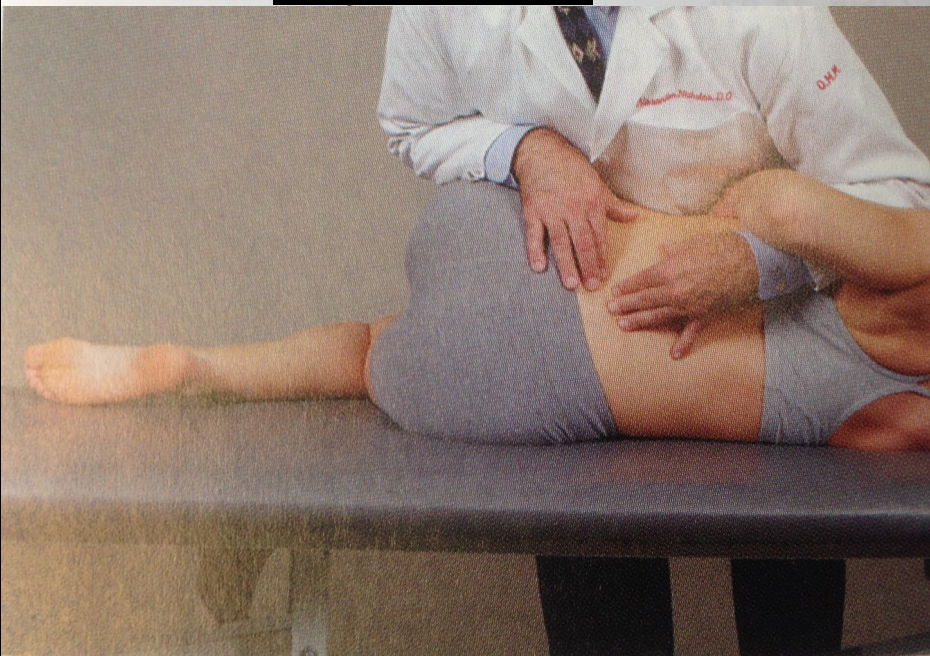


FIGURE 11.92.

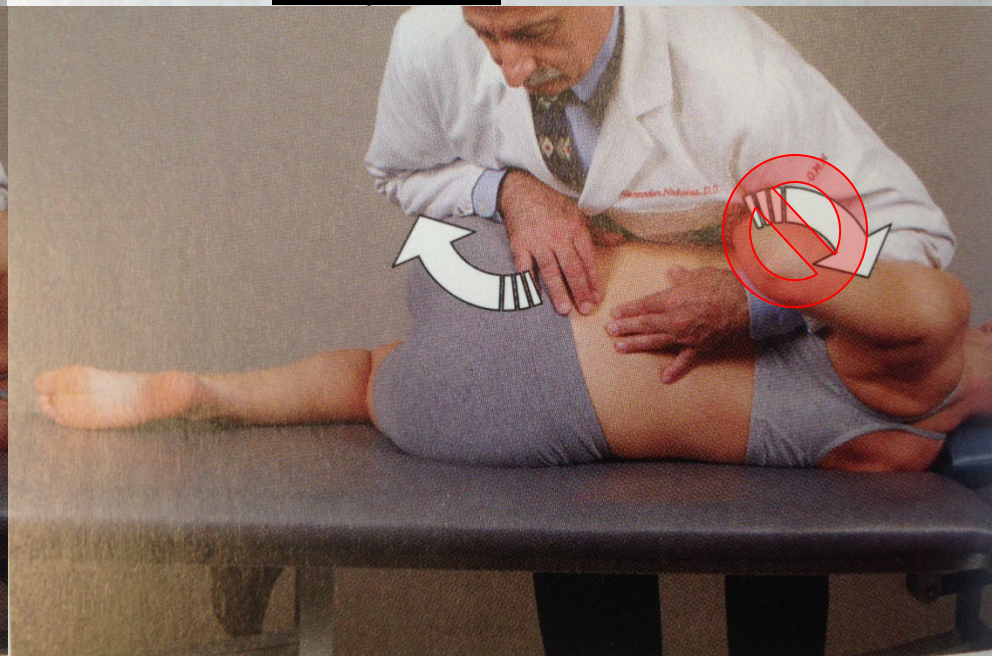


FIGURE 11.93.

Classic Treatment Positions

- The following material is based on classical Facilitated Positional Release applied to the lumbar spine with the patient in a prone position
- There are challenges of frequently needing multiple pillows to produce a neutral spine position and the biomechanics of positioning the patient's lower extremity at the side or under the table
- This information is included for completeness

Extended segment FPR Lumbar Spine L4 ER_L S_L

- See An Osteopathic Approach to Diagnosis and Treatment, 3rd edition; Figure 51-2, page 255 for clearer image



Flexed Segment FPR Lumbar Spine $L4\ FR_L S_L$

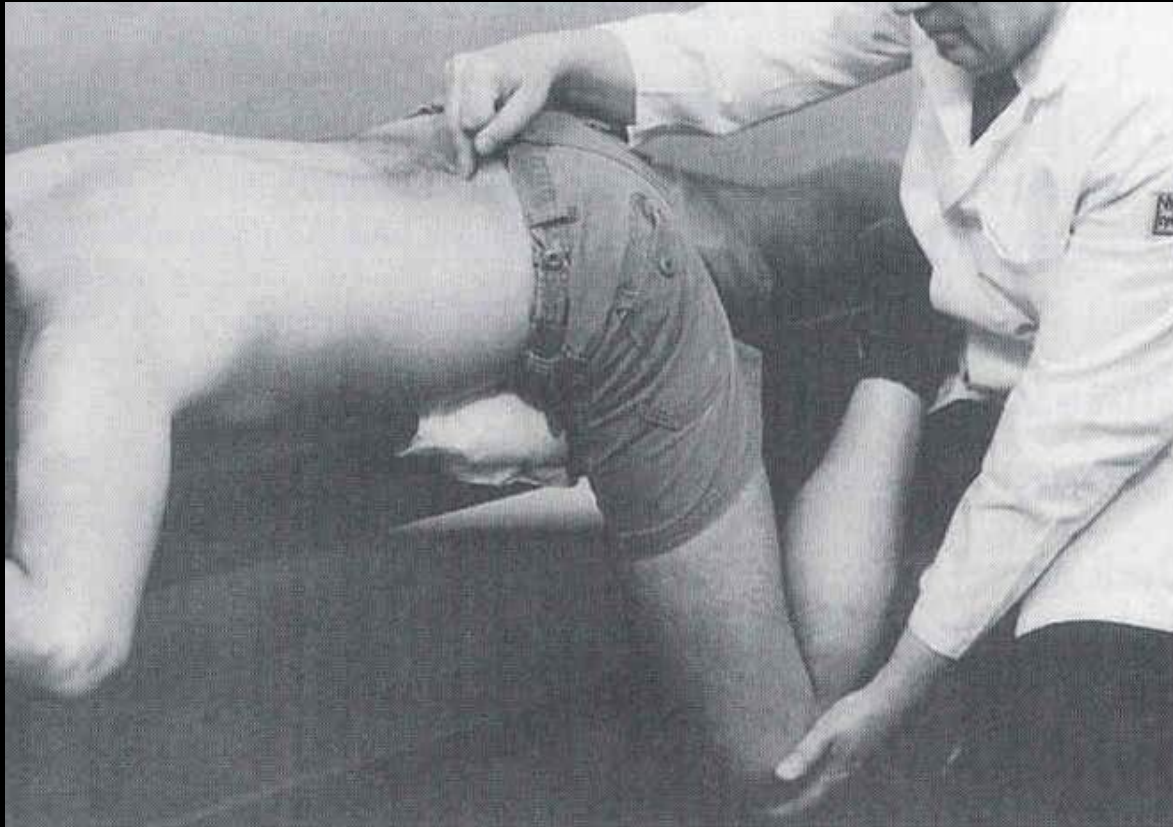


FIGURE 64.8. Lumbar flexed L4 FSLRL technique.

Reading Assignment

An Osteopathic Approach to Diagnosis and Treatment, 4th edition; DiGiovanna, Schiowitz, and Dowling, Ch 15 (pp 72-73); Review index for location based FPR techniques.

Foundations of Osteopathic Medicine, 4th edition; Ch 40G (pp 990-997).